

Penny (ثراء)

The Offline-First Intelligent Financial Companion

Penny is a premium, secure, and offline-first personal financial manager built to empower users with secure, encrypted intelligence.

- **Privacy First:** 100% offline data encryption. No personal data ever leaves the device.
- **On-Device NLP:** Built-in Naive Bayes classification and Trie-based search engines.
- **Bilingual Interface:** Arabic and English options with intelligent language query detection.

NOTE: Designed for resource-constrained environments (like mobile/offline apps) without compromising on advanced AI tools.

The Core Challenge & Our Vision

Rethinking Personal Finance Security

Modern financial apps rely heavily on cloud APIs, exposing user privacy and failing when internet access is unavailable.

Our Solution:

- **Offline Ledger:** Express & SQLite database backend (`karam.db`) running fully inside a local sandbox.
- **Hybrid NLP:** Natural language parsing that resolves intents locally in under **10 microseconds**.
- **Decision Simulator:** Advanced what-if modeling tool to check risk margin levels, account impact, and immediate available cash before committing to real-world expenses.

Key Features & Capabilities

Bilingual Offline Intelligence

Penny handles day-to-day transaction management alongside complex investment modeling.

- **Decision Simulator:** Simulates upfront cost impacts and runs monthly savings risk forecasts.
- **Smart SMS Parser:** Paste SMS notifications from AlRajhi, SNB, STC Pay, SABB, or Riyad Bank to auto-log transactions.
- **Shariah Stock Screener:** Query compliance ratios (debt, impure income) for Saudi stocks (STC, Aramco, Rajhi).
- **Gamified Achievements:** Unlock badges like *Tuwaig Peak* or *Cost Slayer* based on active savings metrics.
- **Auto-Post Salary:** Automated recurring ledger engine that records monthly income on set dates.

System Architecture

Offline-First Integration Flow

Penny is structured to process everything on-device, keeping data processing fast and local.

- **React Client:** Styled with modern glassmorphic aesthetics and responsive flex components.
- **Express API Server:** Integrates NLU intents and SQLite data queries.
- **SQLite Database:** Persistent storage for transactions, subscriptions, goals, stocks, profile, and badges.
- **On-Device ML:** Naive Bayes merchant classifiers and Z-Score anomaly detectors.

High-Performance Offline NLU

Prefix Trees & Synonym Stemming

To ensure responsiveness on mobile platforms, we developed a Custom Stemming & synonym mapping Token Trie structure.

- **Vocabulary Coverage:** Seeds for 20,000+ general inquiries and 10,800 balanced merchants.
- **Response Time:** Sub-millisecond execution averages **0.0058 milliseconds** per query.
- **Fails:** 0 fails across 5,400+ combinatorial queries.

```
--- STARTING NLU DYNAMIC STRESS TESTS --- Generated  
5,400 unique query variations. Average NLU  
classification response time: 0.0058ms Fail count: 0   
NLU Trie Stress Test passed successfully!
```

Premium UX & Design Aesthetics

State-of-the-Art Financial Interface

Penny boasts a premium, high-fidelity dark-themed dashboard designed for maximum engagement.

- **Glassmorphism Panels:** Modern card borders, custom scrollbars, and glowing indicators.
- **LockScreen PIN Overlay:** Secure, animated banking-style lock screen featuring automated focus shifting and lockout penalty countdowns.
- **Auto-Sync Open Banking:** Authentic open banking authentication card simulating SAMA framework policies.
- **Toast Notifications:** Smooth Context-based animated warning/success toasts.